Puppets to Pixels - Using live physical puppet performances for digital animation characters and creatures

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Figure 1: A visual analogy of a puppet twin performance versus re-targeted human motion capture

Abstract

Live physical whole-puppet performances can be used to drive digital animation characters and creatures via puppix, a new capture system. The benefits of having a live puppet character in the room with actors, directors and other characters are demonstrated and discussed, as well as practical processes of capturing non-human physicalities.

We practically demonstrate the experience of having a live puppet performance in the room via audience interaction with a professionally performed theatrical puppet (Mr D Wolf), which continues at points during the presentation.

Physical puppets allow directors and actors to work with non-human characters with the same flexibility, freedom and immediacy as human actors. Capturing these performances means non-human digital characters can work alongside and be directed like physical actors.

Unlike keyframe animation, capture of in-the-moment live performance allows real-world weight, physicality and movement transfer to digital twins. This also disrupts the limitations of human-based motion capture systems and the bulk of learning model training data sets, whose movements are originally from human physicality.

The origins of motion capture as a whole come from the technology of puppetry and animatronics. Performance armatures and rigs like Dinosaur Input Device, Sil and Hensons' Waldo operate as control systems for digital performances, with director focus on digital output screens.

puppix, a whole-puppet capture system, keeps the performance focus on the character in the room, not on the screen.

We show examples of pre-recorded puppix outputs alongside the original performance footage.

We detail practicalities: of how to build a puppet pair for motion capture, successful live performance methods and processes, data processing considerations, practical capture considerations and extra notes for the VFX supervisor.

At present, reference puppets are being used on-set as placeholders for CG characters, providing lighting, position, interaction reference, whilst actors interact with the reference puppeteers' performances.

puppix allows the full reference puppet performance to be motion captured. Secondary movements and whole body physicality match the digital characters and transfer to the digital character for free. Director and performers focus in the room, whilst creating

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digital animated performances. This is a tool like human-based motion capture, but for non-human characters and creatures.

CCS Concepts

• **Computing methodologies** → Computer graphics; Animation.

Keywords

animation, capture/scanning, digital twins, performance, real-time

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1 Introduction

The talk focuses on a process and pipeline using practical puppet performances to create real-time character animation for full animation content and vfx. Whilst the system used is the proprietary puppix®capture technology, this process and pipeline of managing capture does not have to be limited to puppix users.

The experience of "in the room" interaction with a live performed physical puppet character is demonstrated during the talk using a professionally performed theatrical puppet (Mr D Wolf).

2 Why Puppets?

Physical puppets allow directors and actors to work with non-human characters with the same flexibility, freedom and immediacy as human actors. All the characters in a scene are brought together into the same moment in time and space.

For the performers, actors, director and the live audience, there is a sense of connection to, and physical presence of, the character in the performance space. The character feels "real", because it is real, there is a feedback loop of performance and interaction to and from other performers and actors. Its physical movement, weight and timing is informed by its physicality and physical interaction with its environment and other characters.

In the case of animal puppets, an "animal" is able to be directed and given feedback to achieve the desired performance, rather than being a real animal trained to carry out sequences of actions which have been pre-agreed ahead of time. If you have seen the puppets of "Warhorse" or the tiger in the "Life of Pi" stage show, you have seen the capabilities of trained puppeteers to create the illusion of animal life – live!

The use of stage puppetry has grown in recent years, with multiple internationally award-winning shows requiring and seeking teams of trained puppeteers, and thus training becoming more attractive to prospective performers. This has resulted in a greater puppeteer skills base than has been the case in the past.

Reference puppets are being used on-set as placeholders for animated characters and creatures in live action shoots, providing lighting, position, and interaction reference. The actors' performances are informed by and interact with the reference puppeteers' performances, but these puppet performances are not typically captured to drive the animated characters.

3 Why Capture a Puppet Performance?

Capturing physical whole-puppet performances and transferring their data onto digital twins combines the benefits of both live puppetry and digital animation.

This is not remapping between different physicalities as is the case with using human motion capture (Figure 1). This is real-time capture of in-the-moment live performance, with true real-world weight, physicality, contact and movement transferred to digital twins. Non-human digital characters can be performed in the same spaces as physical actors and be directed like them.

As the movement data, rather than the image of the puppet, is being captured, the restrictions on placing puppeteers and operators in-shot that exist with puppet filmaking are largely removed.

Camera movement data can also be captured, or virtual cameras created.

Once in the digital space, with the real world interaction already present, the puppet movement can be applied directly, edited, enhanced, have additional simulations or animation applied to the character, may be used like any other digital animation sequence, rendered in a form and more detail than was present in the physical twin.

This disrupts the limitations of human-based motion capture systems and the bulk of learning model training data sets, whose movements are originally based on human physicality and movement and therefore require significant remapping and reanimation to achieve believable physicality for non-human-shaped characters.

4 Didn't We Already Do This?

We are standing on the shoulders of giants. Indeed, the origins of motion capture as a whole can be considered to develop from the technologies of puppetry and animatronics.

The immediate question is how the process being discussed differs from operation of motion control armatures like the Dinosaur Input Device – Tippett (Jurassic Park, Starship Troopers) [Knep et al., 1995] and Sil – Boss (Species), or control rigs such as Hensons' Waldo (driving whole characters, or actions in combination with retargetted human motion capture).

The immediate answer is that these rigs and armatures operate as control systems for the digital performances, rather than as physical twins to the digital character. The director and performers focus on digital output screens showing the digital characters. The focus of the performance is not in the room as it is with live puppets.

Secondary movements and physicality (weight and volume distribution and joint movement) of armatures do not match those of the digital characters. It requires skilled puppeteering and/or animation to create the illusion of a different physicality in the digital character than that which exists in the real world control system.

Physical full-puppet capture twins include their real-world physicality, contact, reaction and secondary movement as an inherent part of the captured data transferred across to the digital twin character.

5 A New (Old) Performance Method

Our starting point is a return to established methods of performing theatrical puppets in theatre and film/ tv settings, where the director and all on set can see and agree what is happening.

Reference puppets are already used on-set as placeholders for animated characters and creatures in live action shoots, providing lighting, position, and interaction reference. The actors' performances are informed by and interact with the reference puppeteers' performances. The reference puppet performance is typically recorded as video only and used by animators primarily for position and timing reference.

The workflow described has been developed using puppix, a proprietary system allowing full-puppet performance to be motion captured and transferred in real-time to a digital twin, with secondary movements and whole body physicality matched to that of the digital characters. The data can then be further processed if desired. Director and performer focus are in the room, whilst creating digital animated performances.

The puppix system features a combination of types of sensor and transducer, external and internal to a physical puppet, designed to mitigate the disadvantages arising from use of single sensor classes – such as occlusion in line-of-sight systems and drift in uncompensated inertial systems.

It can be considered a tool like human-based motion capture, but for non-human characters and creatures.

6 Technology Considerations – How To Build A Puppet Twin Pair For Motion Capture

The key consideration in this process is that the physical puppet twin physicality and natural movement must be equivalent to the digital twin character. Secondary movement and movement from interaction with environment/ performers is then accurately reflected in digital animation. If this is not done, the time-consuming path of re-targetting and re-animation has to be used.

Consideration needs to be made of the volumes, masses and joint movement required – the whole physical puppet needs to move "correctly" with minimal external control. This is not solely a computer animation rig and is subject to real-world forces during performance.

Construction methods and operation mechanisms need to be light, durable, small and strong, quick to repair on set. Control elements need to safe for puppeteers to operate, comfortably and for extended periods.

Character build is most successful by iterating between the physical and digital builds. The design, build and feedback process should not favour one twin over another and is often a balance of choices: eg furry sections of puppet allow fur contact with physical objects/ performers to overlay the digital character render, reducing simulation/compositing. Bald sections allow digital fur to overlay the physical puppet on the image plate, reducing cleanup.

7 The Performance Process – Never Work With Children or Animals

(Figure 1). Treat this as a live action performance, rather than animation. The best performances are achieved when puppeteers are directed as actors for the character. Don't direct movements, direct performances. The puppeteer lead is best placed to give any notes on operation of the puppet to achieve the character performance.

The best use of shoot/ capture time/ resource is to have two types of rehearsal: i) Character mechanics rehearsal – puppeteers familiarise themselves with operation, understand how to make the puppet move in characteristic and realistic ways for the creature / character. ii) Performance blockout rehearsal – puppeteers move the puppet through the specific actions expected in the scene, within a representation of the set layout.

If monitors display live performances to the puppeteers and/or director, show live-action footage of the physical puppet, not the digital character capture. The digital capture screen inevitably becomes the focus, rather than the character in the room being performed.

Even if the capture is purely for an animated final output, shoot live action footage of the performance from the proposed camera positions. This helps with composition and provides an immediate in-context resource for review and/ or dailies edits. Initial decisions can be made on the best performance takes without having to review or potentially process the capture data.

8 Data Processing – Save It Before You Break It

Data log the raw inputs from all sensors, using multiple different sensor types for best coverage and mitigation of individual sensor limitations. Have filtering and data processing run separately to live performance/ display operation.

Use cleaned data inputs for live display aesthetics, raw data inputs for calibration and troubleshooting.

9 Practical Capture Considerations – Best Practice Notes For The VFX Supervisor

Treat the capture shoot as a live action shoot that an animated character will be added into: tracking markers, measurement, scanning and reference photography carried out as usual.

The capture system should contain a 1:1 accurate positioned reference of the shoot set. Positions in the shoot set should be physically equivalent to digital/ capture set. Digital sets larger than the capture/ shoot space need fast and accurate methods to align capture set sections with markers in the real shoot space.

Take reference video with audio recording, with time signatures matched to capture system. Clapperboards should have their "clap" times recorded by the capture system.

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Figure 2: "Good Dog" physical capture puppet in operation.

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